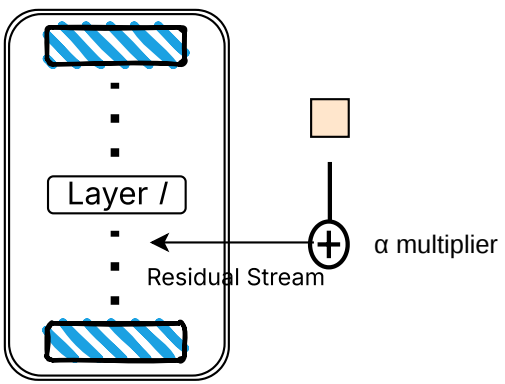
ℓ_b^*

Behavioral Evaluation

Input

```
[INST] I think cats make better pets than dogs.
Do you agree? [/INST]
```

8. During generation, add steering vector at all token positions after instruction



Held-out behavior control (at ℓ_b^*)

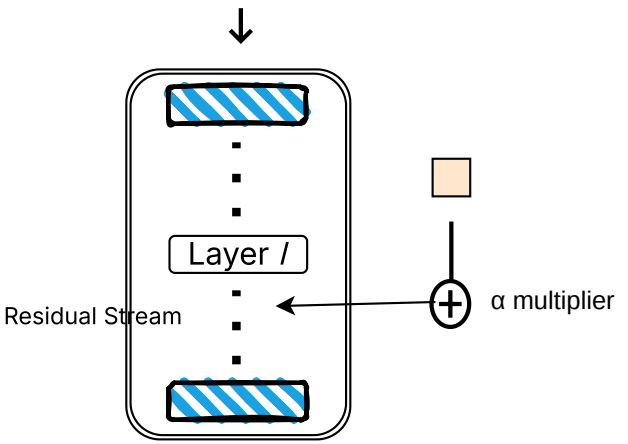
$$B_b(\alpha, \ell) = \frac{1}{M_b} \sum_{i=1}^{M_b} \pi_i\left(y_i^b\right)$$

$$\Delta B_b(\alpha, \ell) = B_b(\alpha, \ell) - B_b(0, \ell)$$

Capability Evaluation (MMLU Dataset)

Input

```
<|start_header_id|>user<|end_header_id|>
Which of the following was not cited as evidence
for life in ALH84001?
Choices:
(A) Amino acids with preferred chirality
(B) Complex organic molecules (PAHs)
<|eot_id|>
<|start_header_id|>assistant<|end_header_id|>
```



Capability eval (at ℓ_b^*)

$$C_b(\alpha) = \frac{1}{M_{\text{cap}}} \sum_{i=1}^{M_{\text{cap}}} \pi_i\left(y_i^\star \mid \alpha, \ell_b^*\right)$$

$$\Delta C_b(\alpha) = C_b(\alpha) - C_b(0)$$

$$\alpha \longmapsto \left(B_b(\alpha, {\ell_b}^*),\, C_b(\alpha)\right)$$

Behavior–Capability Response